SQL — INSERT, UPDATE, DELETE & JOINS

The Master Librarian - Managing & Connecting Data

HOW DOES ZOMATO KNOW YOUR ORDER?

It's All About Joining Information!

How does the Zomato delivery person know your name AND your order? The answer lies in connecting different pieces of information stored in databases!

The Power of Joins

RECAP: TABLES & PRIMARY KEYS

Remember, a table is like a grid with rows and columns — just like your class attendance register! Each row is one record, and each column stores one type of information.

A Primary Key is a unique ID for each row — like your Roll Number!

Example: Students table has RollNo, Name, and Class columns.

No two students can have the same Roll Number — that's what makes it special!

ADDING NEW RECORDS WITH INSERT

Want to add a new player to the cricket team? The INSERT command lets you add new rows of data to any table. It's like writing a new entry in your register!

INSERT INTO Players (PlayerID, Name, Runs) VALUES (7, 'Virat', 0);

Before: Table has 6 players listed

After: Table now shows 7 players including Virat!

FIXING DATA WITH UPDATE

The UPDATE command lets you change existing data in your table. Always use WHERE to specify which row to update — otherwise all rows change!

```
UPDATE Players SET Runs = 75 WHERE PlayerID = 7;
```

Before: Virat had 0 runs in the table.

After: Virat now shows 75 runs scored!

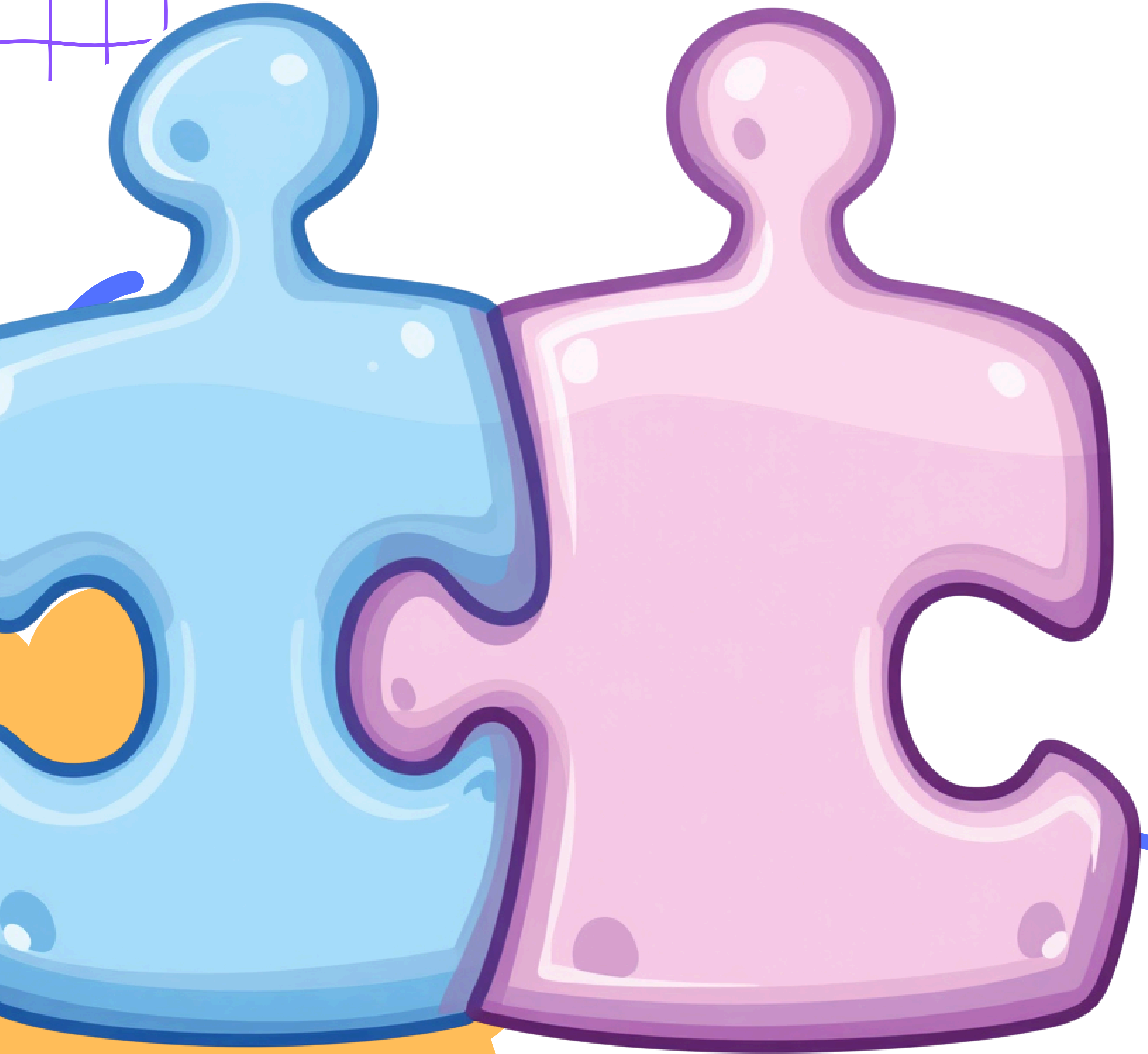
REMOVING RECORDS WITH DELETE

Sometimes players retire or leave the team. The DELETE command removes records from your table. Always use WHERE to specify exactly which record to remove!

```
DELETE FROM Players WHERE PlayerID = 3;
```

Before: Table shows retired player in row 3

After: Player removed, table updated!



Section Divider

CONNECTING DATA WITH JOINS

The Link that Brings Tables Together

WHAT IS A JOIN?

The Puzzle Piece Connection

JOIN = Connecting two tables using a common key. Think of it like puzzle pieces fitting together! The Students table and Exam Scores table link through Roll Number.

JOIN connects student names to their scores!

VISUALIZING TWO TABLES

See how two tables can share a common column? The Roll No appears in both tables — this is the key that links them together!

Students Table: Roll No, Name, Class

Library Table: Roll No, Book Title, Return Date

JOIN IN ACTION: WHO HAS 'PANCHATANTRA'?

```
SELECT Students.Name, Library.BookTitle FROM Students
JOIN Library ON Students.RollNo = Library.RollNo WHERE
Library.BookTitle = 'Panchatantra';
```

The JOIN connects both tables using RollNo as the link.

Result shows: Priya borrowed 'Panchatantra' — matched rows highlighted!

JOINs help us answer questions across multiple tables.

DATABASE MANAGER CHALLENGE

Can You Pick the Right Command?

Read each scenario carefully, then choose the correct SQL command: INSERT, UPDATE, DELETE, or JOIN. Think like a master librarian!

Time to test your SQL skills!

QUIZ 1: NEW ADMISSION

Scenario: A new student named Priya has joined Class 6B. You need to add her information to the school database. What command should you use?

WHICH SQL COMMAND WILL YOU USE?

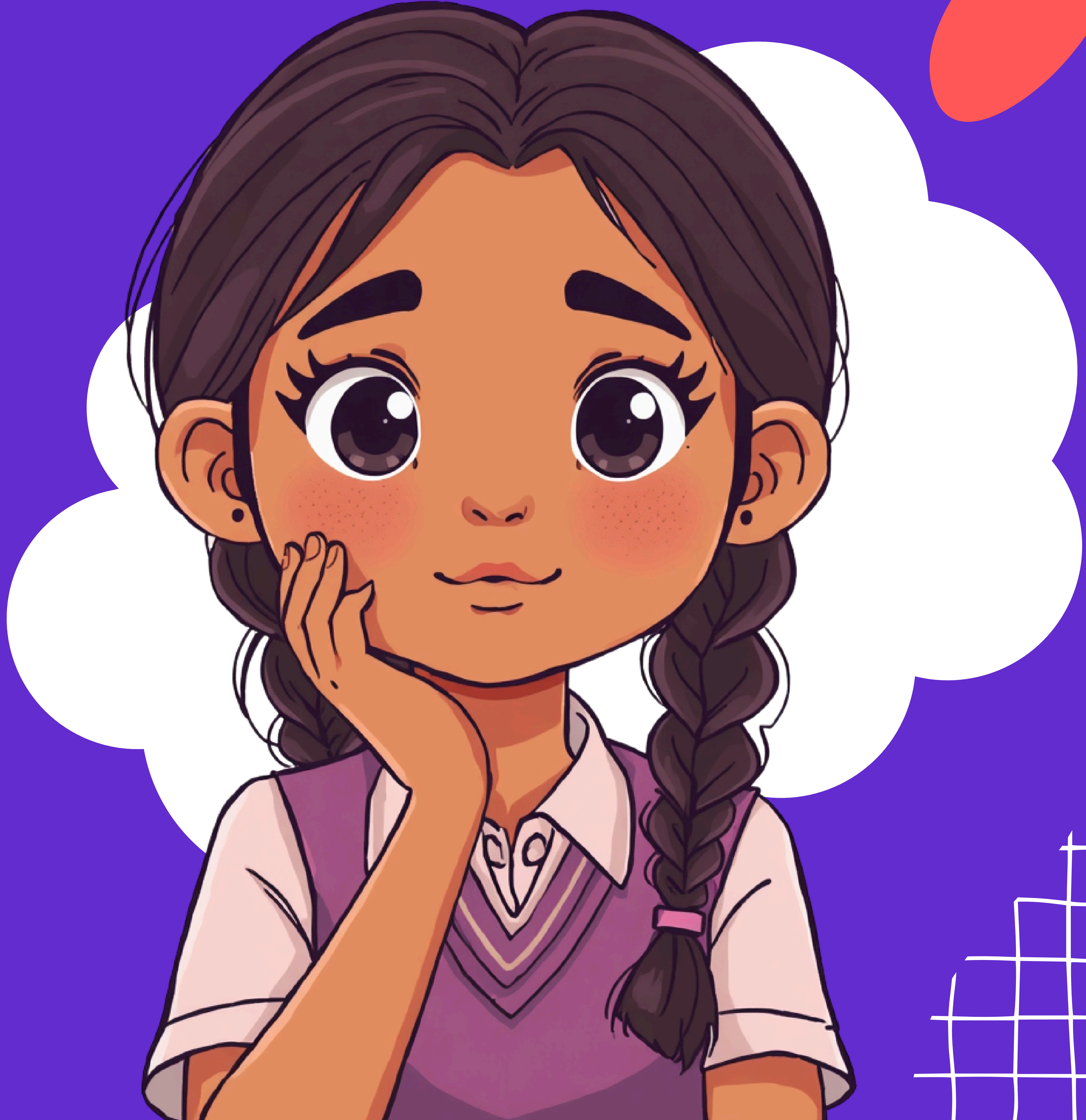
INSERT | UPDATE | DELETE | JOIN

QUIZ 2: ADDRESS CHANGE

Scenario: A student named Priya has moved to a new house. You need to change her address in the school database. Which SQL command will help you fix this?

WHICH COMMAND WOULD YOU USE?

INSERT / UPDATE / DELETE / JOIN



QUIZ 3: LOST LIBRARY BOOK

Scenario: "Remove a lost book from the library database."
Options:
- INSERT
- UPDATE
- DELETE
- JOIN

WHICH SQL COMMAND WOULD YOU USE?

QUIZ 4: FIND EXAM SCORES

Scenario: "See which student scored what in exams."
A) INSERT
B) UPDATE
C) DELETE
D) JOIN

WHICH COMMAND WOULD YOU USE?

QUIZ 5: ADD A NEW CRICKET PLAYER

Scenario: "Add a new cricket player to the team."
Think carefully — is this about adding, changing, removing, or connecting data?

WHICH COMMAND WOULD YOU USE?

INSERT / UPDATE / DELETE / JOIN

THANK YOU!

Great job! You are now the Master Librarian! You have learned how to INSERT new records, UPDATE existing data, DELETE unwanted entries, and JOIN tables together. These powerful SQL commands put you in control of the data!

Keep practicing your SQL magic! See you in the next session at Planrs Academy.